

Limits and Continuity

CSC 1105 Mathematics for Computer Science

Denish Azamuke, PhD

Office: CoCIS, Block B, Level 3

August 17, 2026

Module 1 at a Glance

Module

Limits and Continuity

- Limits of simple and composite functions
- Limit laws and standard techniques
- Limits to infinity and horizontal asymptotes
- Continuity and differentiability tests
- Derivatives and chain rule applications

Why it matters in computing

- Algorithm growth rates
- Machine learning gradients
- Computer graphics curves
- Signal and image processing
- Performance modeling

Learning Outcomes

By the end of this module, students should be able to:

- 1 Explain the meaning of a limit using numerical, graphical, and algebraic views.
- 2 Compute limits of polynomial, rational, trigonometric, exponential, logarithmic, and mixed functions.
- 3 Use one-sided limits to determine whether a two-sided limit exists.
- 4 Test continuity of functions at points and on intervals.
- 5 Relate differentiability to continuity.
- 6 Apply derivatives to rates of change and simple optimization problems.

Functions

Definition

A function assigns each input x from a domain exactly one output $f(x)$.

Examples:

$$f(x) = x^2 - 4, \quad g(x) = \frac{x^2 - 1}{x - 1}, \quad h(x) = e^x \sin x.$$

- A limit asks: what value does $f(x)$ approach as x approaches a point?
- The function value at the point may or may not be defined.

The Intuitive Meaning of a Limit

Notation

$$\lim_{x \rightarrow a} f(x) = L$$

means that $f(x)$ gets arbitrarily close to L as x gets close to a .

- The symbol $x \rightarrow a$ means x approaches a , not necessarily equals a .
- The symbol L is the value being approached.
- Limits describe behavior near a point.

Numerical View of a Limit

Consider

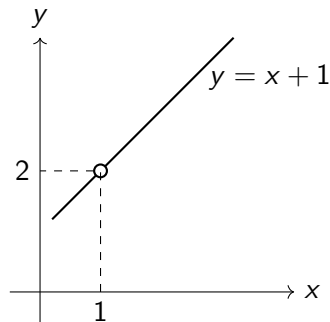
$$f(x) = \frac{x^2 - 1}{x - 1}, \quad x \neq 1.$$

x	0.9	0.99	0.999	1.001	1.01
$f(x)$	1.9	1.99	1.999	2.001	2.01

Even though $f(1)$ is undefined, the values approach 2. Therefore,

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2.$$

Graphical View: Hole but Limit Exists



$$f(x) = \frac{x^2 - 1}{x - 1}, \quad x \neq 1.$$

$$\lim_{x \rightarrow 1} f(x) = 2$$

but $f(1)$ is undefined.

One-Sided Limits

Left-hand limit

$$\lim_{x \rightarrow a^-} f(x)$$

Approach a using values less than a .

Right-hand limit

$$\lim_{x \rightarrow a^+} f(x)$$

Approach a using values greater than a .

Two-sided limit test

$$\lim_{x \rightarrow a} f(x) = L \text{ exists if and only if } \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L.$$

Example: One-Sided Limits

Let

$$f(x) = \begin{cases} x + 2, & x < 1, \\ 4, & x = 1, \\ 3x, & x > 1. \end{cases}$$

$$\lim_{x \rightarrow 1^-} f(x) = 3, \quad \lim_{x \rightarrow 1^+} f(x) = 3.$$

Therefore,

$$\lim_{x \rightarrow 1} f(x) = 3.$$

Notice that $f(1) = 4$, but the limit is 3.

Limit Laws

If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then:

$$\lim_{x \rightarrow a} [f(x) + g(x)] = L + M,$$

$$\lim_{x \rightarrow a} [f(x) - g(x)] = L - M,$$

$$\lim_{x \rightarrow a} [cf(x)] = cL,$$

$$\lim_{x \rightarrow a} [f(x)g(x)] = LM,$$

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad M \neq 0.$$

Direct Substitution

For many continuous elementary functions, try direct substitution first.

Example

$$\lim_{x \rightarrow 2} (3x^2 - 5x + 4) = 3(2)^2 - 5(2) + 4 = 6.$$

- Direct substitution works for polynomials everywhere.
- It also works for rational functions where the denominator is not zero.
- If substitution gives $0/0$, more work is needed.

Indeterminate Form $0/0$

The expression $0/0$ does not mean the limit is zero or undefined immediately. It means the method is inconclusive.

Common techniques

- Factor and cancel
- Rationalize using conjugates
- Use standard limits
- Use the squeeze theorem
- Use L'Hopital's rule only after derivatives have been introduced

Technique 1: Factor and Cancel

Example

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = \lim_{x \rightarrow 3} \frac{(x - 3)(x + 3)}{x - 3} = \lim_{x \rightarrow 3} (x + 3) = 6.$$

- Cancellation is valid for $x \neq 3$.
- **Warning!**
The limit depends on nearby values, not only the value at $x = 3$.

Technique 2: Rationalize

Example

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$$

Multiply by the conjugate:

$$\begin{aligned} \frac{\sqrt{x+4} - 2}{x} \cdot \frac{\sqrt{x+4} + 2}{\sqrt{x+4} + 2} &= \frac{x+4-4}{x(\sqrt{x+4} + 2)} \\ &= \frac{1}{\sqrt{x+4} + 2}. \end{aligned}$$

Thus,

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x} = \frac{1}{4}.$$

Special Trigonometric Limits

Two important standard limits are:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0.$$

Example

$$\lim_{x \rightarrow 0} \frac{\sin(5x)}{x} = 5 \lim_{x \rightarrow 0} \frac{\sin(5x)}{5x} = 5.$$

The Squeeze Theorem

Statement

If

$$g(x) \leq f(x) \leq h(x)$$

near a , and

$$\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = L,$$

then

$$\lim_{x \rightarrow a} f(x) = L.$$

Classic example

$$\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0.$$

Squeeze Theorem: Example

Evaluate

$$\lim_{x \rightarrow 0} x^2 \sin \left(\frac{1}{x} \right).$$

Since

$$-1 \leq \sin \left(\frac{1}{x} \right) \leq 1,$$

and $x^2 \geq 0$, multiplying by x^2 gives

$$-x^2 \leq x^2 \sin \left(\frac{1}{x} \right) \leq x^2.$$

But

$$\lim_{x \rightarrow 0} (-x^2) = 0 \quad \text{and} \quad \lim_{x \rightarrow 0} x^2 = 0.$$

Therefore, by the Squeeze Theorem,

$$\boxed{\lim_{x \rightarrow 0} x^2 \sin \left(\frac{1}{x} \right) = 0}.$$

L'Hopital's Rule

Suppose

$$\lim_{x \rightarrow a} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 0.$$

Then the limit

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

has the indeterminate form $0/0$.

L'Hopital's Rule

If f and g are differentiable near a , and $g'(x) \neq 0$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)},$$

provided the limit on the right exists.

L'Hopital's Rule: Worked Example

Evaluate

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x}.$$

First check the form:

$$\frac{e^0 - 1}{0} = \frac{0}{0}.$$

So we may apply L'Hopital's Rule.

Differentiate the numerator and denominator:

$$\frac{d}{dx}(e^x - 1) = e^x, \quad \frac{d}{dx}(x) = 1.$$

Therefore,

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \lim_{x \rightarrow 0} \frac{e^x}{1} = e^0 = \boxed{1}.$$

Limits at Infinity

Meaning

$$\lim_{x \rightarrow \infty} f(x) = L$$

means that $f(x)$ approaches L as x becomes very large.

For rational functions, compare highest powers.

Example

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 2x - 1}{x^2 - 5} = 3.$$

Limits at Infinity: Worked Example

Evaluate

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 2x - 1}{x^2 - 5}.$$

Divide numerator and denominator by the highest power of x , which is x^2 :

$$\frac{3x^2 + 2x - 1}{x^2 - 5} = \frac{3 + \frac{2}{x} - \frac{1}{x^2}}{1 - \frac{5}{x^2}}.$$

As $x \rightarrow \infty$,

$$\frac{2}{x} \rightarrow 0, \quad \frac{1}{x^2} \rightarrow 0, \quad \frac{5}{x^2} \rightarrow 0.$$

Therefore,

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 2x - 1}{x^2 - 5} = \frac{3 + 0 - 0}{1 - 0} = \boxed{3}.$$

Horizontal Asymptotes

A horizontal asymptote is a line $y = L$ that a graph approaches as $x \rightarrow \infty$ or $x \rightarrow -\infty$.

Example

For

$$f(x) = \frac{2x + 1}{x - 4},$$

we have

$$\lim_{x \rightarrow \infty} f(x) = 2.$$

So $y = 2$ is a horizontal asymptote.

Exponential and Logarithmic Limits

- Exponential functions grow very quickly: a^x , where $a > 1$.
- Logarithmic functions grow slowly: $\ln x$.
- Both appear in computer science, especially in algorithm analysis and information theory.

Examples

$$\lim_{x \rightarrow \infty} e^{-x} = 0, \quad \lim_{x \rightarrow \infty} \frac{\ln x}{x} = 0.$$

Continuity: The Basic Idea

Idea

A function is **continuous at** $x = a$ if there is no break, hole, or jump in the graph at $x = a$.

In simple terms, as x gets closer to a , the values of $f(x)$ must get closer to the actual value $f(a)$.

Thus, f is continuous at $x = a$ if:

- 1 $f(a)$ is defined,
- 2 $\lim_{x \rightarrow a} f(x)$ exists,
- 3 $\lim_{x \rightarrow a} f(x) = f(a)$.

In short:

The limit at a must equal the function value at a .

Testing Continuity

To test whether f is continuous at a :

- 1 Compute $f(a)$.
- 2 Compute $\lim_{x \rightarrow a^-} f(x)$.
- 3 Compute $\lim_{x \rightarrow a^+} f(x)$.
- 4 Check whether all three values are equal.

Key condition

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a).$$

Testing Continuity: Example 1

Test whether

$$f(x) = \begin{cases} x + 2, & x < 1, \\ 3, & x = 1, \\ 2x + 1, & x > 1 \end{cases}$$

is continuous at $x = 1$.

$$f(1) = 3$$

$$\lim_{x \rightarrow 1^-} f(x) = 1 + 2 = 3$$

$$\lim_{x \rightarrow 1^+} f(x) = 2(1) + 1 = 3$$

Since

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = f(1) = 3,$$

f is continuous at $x = 1$.

Types of Discontinuity

Removable

A hole exists, but the limit exists.

Jump

Left and right limits are finite but unequal.

Infinite

The function grows without bound near a point.

Identifying discontinuities helps us understand system behavior, numerical instability, and model boundaries.

Continuity: Example 2

Let

$$f(x) = \begin{cases} x^2 + 1, & x < 2, \\ kx, & x \geq 2. \end{cases}$$

Find k so that f is continuous at $x = 2$.

$$\begin{aligned} \lim_{x \rightarrow 2^-} f(x) &= 2^2 + 1 = 5, \\ f(2) &= 2k. \end{aligned}$$

For continuity: $2k = 5$. Therefore,

$$k = \frac{5}{2}.$$

Intermediate Value Theorem (IVT)

Theorem

If f is continuous on $[a, b]$, and N is between $f(a)$ and $f(b)$, then there is at least one number $c \in (a, b)$ such that

$$f(c) = N.$$

Interpretation

A continuous function cannot jump over values.

Using IVT to Show a Root Exists

Show that $f(x) = x^3 + x - 1$ has a root between 0 and 1.

$$f(0) = -1,$$

$$f(1) = 1.$$

Since f is a polynomial, it is continuous. Because 0 lies between -1 and 1 , the Intermediate Value Theorem guarantees a $c \in (0, 1)$ such that

$$f(c) = 0.$$

IVT: Worked Example 1

Show that

$$f(x) = x^3 + x - 1$$

has a root between 0 and 1.

$$f(0) = 0^3 + 0 - 1 = -1$$

$$f(1) = 1^3 + 1 - 1 = 1$$

So,

$$f(0) < 0 \quad \text{and} \quad f(1) > 0.$$

Since $f(x) = x^3 + x - 1$ is a polynomial, it is continuous on $[0, 1]$.

Therefore, by the Intermediate Value Theorem, there exists some

$$c \in (0, 1)$$

such that

$$f(c) = 0.$$

IVT: Worked Example 2

Show that

$$f(x) = x^3 - 4x + 1$$

has a root between 0 and 1.

$$f(0) = 0^3 - 4(0) + 1 = 1$$

$$f(1) = 1^3 - 4(1) + 1 = -2$$

So,

$$f(0) > 0 \quad \text{and} \quad f(1) < 0.$$

Since $f(x) = x^3 - 4x + 1$ is a polynomial, it is continuous on $[0, 1]$.

Therefore, by the Intermediate Value Theorem, there exists some

$$c \in (0, 1)$$

such that

$$f(c) = 0.$$

Differentiability and Continuity

Important relationship

If f is differentiable at $x = a$, then f is continuous at $x = a$.

But the converse is not always true

A function can be continuous but not differentiable.

Example:

$$f(x) = |x|$$

is continuous at $x = 0$, but it has a sharp corner, so it is not differentiable at $x = 0$.

Differentiability and Continuity: Example

Consider

$$f(x) = |x| = \begin{cases} -x, & x < 0, \\ x, & x \geq 0. \end{cases}$$

Step 1: Check continuity at $x = 0$.

$$\lim_{x \rightarrow 0^-} f(x) = 0, \quad \lim_{x \rightarrow 0^+} f(x) = 0, \quad f(0) = 0.$$

Therefore,

$$\lim_{x \rightarrow 0} f(x) = f(0) = 0,$$

so f is **continuous at $x = 0$** .

Differentiability and Continuity: Example

Step 2: Check differentiability at $x = 0$.

$$f'_-(0) = -1, \quad f'_+(0) = 1.$$

Since

$$f'_-(0) \neq f'_+(0),$$

the derivative at $x = 0$ does not exist.

$f(x) = |x|$ is continuous at 0, but not differentiable at 0.

Derivative as a Limit

Definition

The derivative of f at x is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h},$$

provided the limit exists.

- It measures the instantaneous rate of change.
- It gives the slope of the tangent line.
- It is central to **optimization** and **machine learning**.

Worked Derivative from First Principles

Find the derivative of $f(x) = x^2$.

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} (2x + h) \\ &= 2x. \end{aligned}$$

Basic Derivative Rules

$$\frac{d}{dx}(c) = 0,$$

$$\frac{d}{dx}(x^n) = nx^{n-1},$$

$$\frac{d}{dx}[cf(x)] = cf'(x),$$

$$\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x),$$

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x).$$

The Chain Rule

Rule

If $y = f(g(x))$, then

$$\frac{dy}{dx} = f'(g(x))g'(x).$$

Example

$$y = (3x^2 + 1)^5$$

$$y' = 5(3x^2 + 1)^4(6x) = 30x(3x^2 + 1)^4.$$

Rates of Change

A derivative can describe how fast one quantity changes with respect to another.

Example: position and velocity

If

$$s(t) = t^2 + 3t,$$

then velocity is

$$v(t) = s'(t) = 2t + 3.$$

At $t = 4$,

$$v(4) = 11.$$

Optimization Example

A simple cost model for an algorithm is

$$C(x) = x^2 - 10x + 40.$$

Find the value of x that minimizes cost.

$$\begin{aligned}C'(x) &= 2x - 10, \\2x - 10 &= 0, \\x &= 5.\end{aligned}$$

Since $C''(x) = 2 > 0$, the cost is minimized at $x = 5$.

Computer Science Connections

- **Algorithm analysis:** limits describe long-run growth behavior.
- **Machine learning:** derivatives help optimize loss functions.
- **Graphics:** continuity controls smooth curves and surfaces.
- **Robotics:** rates of change describe velocity and acceleration.
- **Security and networks:** mathematical models use continuous functions to describe trends and thresholds.

Common Mistakes – Warning!

- Assuming $f(a)$ must equal the limit.
- Forgetting to check both one-sided limits.
- Cancelling factors before factoring correctly.
- Treating $0/0$ as a final answer.
- Confusing continuity with differentiability.
- Applying L'Hopital's rule before confirming an indeterminate form.

Revision Questions

For each function, determine the limit and state the method used.

① $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$

② $\lim_{x \rightarrow 0} \frac{\sqrt{x+9} - 3}{x}$

③ $\lim_{x \rightarrow 0} \frac{\sin(7x)}{x}$

④ $\lim_{x \rightarrow \infty} \frac{5x^2 - x + 2}{2x^2 + 3}$

⑤ $\lim_{x \rightarrow 1} \frac{\ln x}{x - 1}$

Hint (methods): factoring, rationalization, standard trigonometric limit, highest-power comparison, and L'Hopital's Rule.

Tutorial Exercises

- 1 Compute $\lim_{x \rightarrow -1} (x^3 + 2x^2 - 5)$.
- 2 Compute $\lim_{x \rightarrow 4} \frac{x - 4}{\sqrt{x} - 2}$.
- 3 Determine whether $f(x) = \frac{x^2 - 1}{x - 1}$ is continuous at $x = 1$.
- 4 Find k so that $f(x) = \begin{cases} kx + 1, & x < 2 \\ x^2, & x \geq 2 \end{cases}$ is continuous at $x = 2$.
- 5 Differentiate $y = (2x^3 - 1)^4$.
- 6 Use IVT to show that $x^3 - 2x - 1 = 0$ has a root in $[1, 2]$.

Quick Self-Check

Answer these before the next class:

- 1 What does $\lim_{x \rightarrow a} f(x) = L$ mean in words?
- 2 When does a two-sided limit fail to exist?
- 3 What three conditions define continuity at a point?
- 4 Why does differentiability imply continuity?
- 5 Give one computer science application of derivatives.

Questions?